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Heterogeneous Material Stress-Strain and Failure Analysis.

Computer vision system for non-contact stress-strain assessment and failure mechanism analysis in tension tests of materials with heterogeneous microstructure

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Abstract

In this contribution we present a tension test system comprised of an open hardware testing machine and a set of open-source computer vision algorithms for non-contact stress-strain assessment.

The hardware is a homemade equipment customized for micro-tension experiments with maximum load of 50 kgf. We present a detailed set of building instructions, where the cost of the materials bill is around 1500 USD, a set of experiments evaluating the machine performance and limitations, mechanical test characterization for a wide set of materials, including those with heterogeneous microstructure distribution as well as reference materials for interlaboratory comparison.

Materials and Methods

System Environments

The overall system was designed to integrate three environments that, when coupled and synchronized, form a unified whole capable of performing deformation analysis and failure mechanism assessments:

- Uniaxial tension generation and hardware control.
- Image acquisition
- Computational control

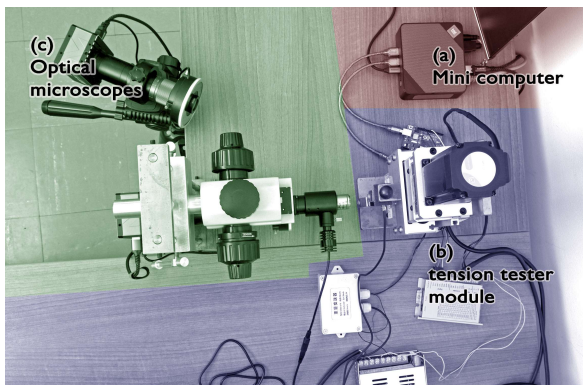


Figure 1: System Environments.

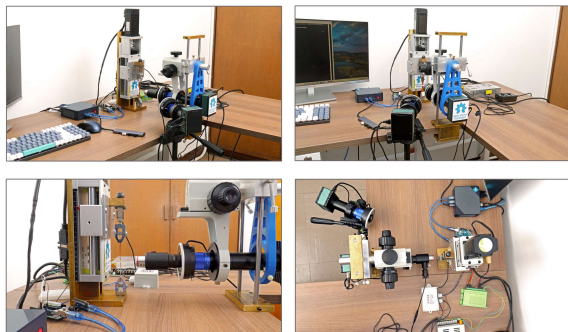


Figure 2: Assay on copper sample.

Results and Analysis



Figure 3: Test specimens: Stainless steel on copper, tissue scaffolds, and biopolymers (from left to right).

Below, Figures 4, 5, and 6 show both the interface operation and the results of deformations and failure mechanisms.

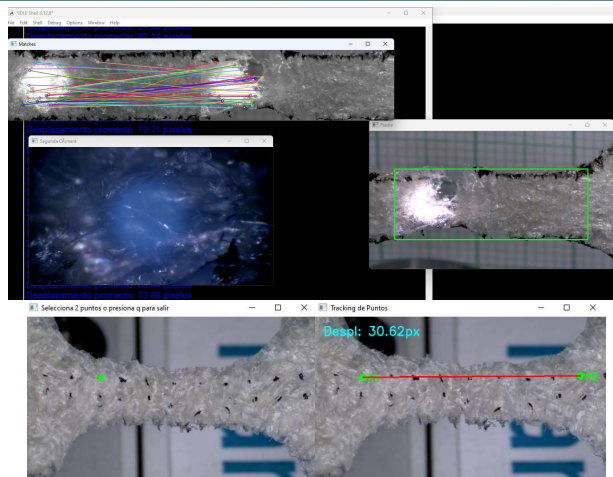


Figure 4: Stress-strain assay in bovine collagen scaffold.



Figure 5: Failure mechanism in copper with steel coating.

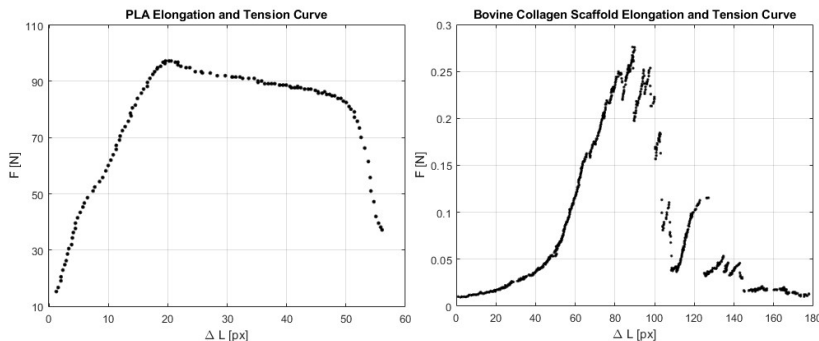


Figure 6: Elongation Tension Curves.

Conclusion

- The prototype is an innovative alternative for the analysis of heterogeneous materials. Its results are promising, though optimization is required to expand its application across different study contexts. It shows strong potential for analyzing deformations and failure mechanisms.

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References

For complete project references, scan the QR code.

